

Math 421

Friday, September 26

Announcements

- Homework 3 due **today**

Additional Consequences of Completeness

Thm (Density of $\mathbb{Q}$). If $a, b \in \mathbb{R}$ and $a < b$, then there is a rational $r \in \mathbb{Q}$ such that $a < r < b$.

Thm (Existence of Square Roots). If $x \geq 0$, then there exists a real number $\alpha \in \mathbb{R}$ satisfying $\alpha \geq 0$ and $\alpha^2 = x$ (i.e. $\sqrt{x}$ exists).

Recipe to Show $\sup S = M$

Step 1: Prove that M is an upper bound for S (i.e show $M \geq s$ for all $s \in S$) and therefore $\sup S$ exists.

Step 2: Prove that M is the least upper bound by using one of the following methods.

- **Method 1:** Assume there exists another upper bound, M_0 , prove that $M < M_0$.
- **Method 2:** Show that for all $\epsilon > 0$, there exists $s \in S$ such that $s > M - \epsilon$.

Example 1: Proof Sketch Using Recipe

Theorem. Assume $A \subseteq [0, \infty)$ is nonempty and bounded above. If $B = \{a^2 : a \in A\}$, then $\sup B = (\sup A)^2$.

Proof Sketch.

Example 2: Proof Sketch Using Recipe

Theorem. Let $A = [a, b)$. Then $\sup A = b$.

Proof Sketch.

Chapter 2: Sequences

Section 7: Limits of Sequences

Sequences

Definition. A **sequence** is a function that has a specified value for each integer $n \geq m$ (usually m is either 0 or 1).

Notation: $(s_n)_{n \in \mathbb{N}}$ or $(s_1, s_2, s_3, \dots)$

Examples:

1. $(s_n)_{n \in \mathbb{N}}$ where $s_n = \frac{1}{n^2}$
2. $(s_n)_{n=0}^{\infty}$ where $s_n = \sin \frac{n\pi}{4}$

Convergence

Def: A sequence (s_n) of real numbers is said to **converge** to the real number s provided that:

For each $\epsilon > 0$ there exists a number N such that
whenever $n > N$ implies $|s_n - s| < \epsilon$.

Activity

Give an example of the following:

1. Two different sequences that have the same *set of values*.
2. Two different sequences that converge to the same real number s .

Group Activity – Divide & Conquer

We can think of the definition of convergence as a game:

someone gives you (a very small) $\epsilon > 0$ and you need to find some N that makes the statement “whenever $n > N$, it follows that $|s_n - s| < \epsilon$ ” true. Find the appropriate N for each of the following examples:

1. $s_n = \frac{1}{n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

2. $s_n = \frac{1}{e^n}, s = 0, \epsilon = 1, \frac{1}{10}, \frac{1}{100}$

Recipe to show that $\lim s_n = s$

Example: Show that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$

Discussion.

Proof.

Example: Show that $\lim \frac{n^2+1}{n^2} = 1$

Discussion.

Activity – Write the Proof.

Activity: Show that $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} = 0$

Discussion.

Proof.